Electrodynamics II, Module 1: study notes

Electric and magnetic fields in matter (Griffiths 5th ed. excerpt, pp. 166–208 and 268–297). Student reading notes in my own words; the book is the reference. Figures are image-generated sketches.

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Section 4.1 · Polarization

Metals and insulators behave differently. In a metal, some electrons are free and roam through the whole piece. In an insulator (also called a dielectric), every electron is tied to its own atom or molecule. So I thought: if the electrons cannot leave, nothing should happen when I put it in a field.

That is wrong, and the book's first point is that the atom can still change shape a little. The electrons can shift slightly away from the nucleus, and that small shift is enough to matter.

2. Induced dipoles: what happens to a neutral atom?

An atom has a nucleus with charge $+q$ and an electron cloud with total charge $-q$. Overall it is neutral. Now put it in a field $\mathbf{E}$ pointing to the right. The nucleus is pushed right, the cloud is pushed left. The nucleus and cloud separate by a small distance d.

Why do they stop? Because the cloud is negative and the nucleus is positive, they pull each other back. When the pull from the cloud equals the push from the field, they stop. That is equilibrium.

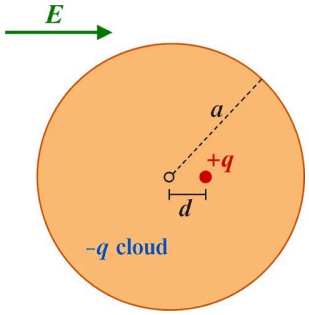


Figure 1. The Example 4.1 model: the nucleus is pushed right by E and the cloud pulls it back.

Finding the pull from the cloud (the model in Example 4.1)

To make the sum possible, the book pictures the cloud as a uniform ball of charge $-q$ and radius a. Now I need the field of that ball at a distance d from its centre, with $d < a$.

Gauss's law says the flux through a sphere of radius s equals the enclosed charge divided by ϵ_0. The fraction of the ball's charge inside radius s is $(s/a)^3$, because the charge is spread evenly through the volume. So the enclosed charge is $q\,s^3/a^3$. The flux through the sphere is $E \cdot 4\pi s^2$, so

$$E \cdot 4\pi s^2 = \frac{q\,s^3}{a^3\,\epsilon_0} \quad \Longrightarrow \quad E_e = \frac{1}{4\pi\epsilon_0}\frac{q\,d}{a^3}.$$

Notice the field grows with d. The further the nucleus moves, the harder the cloud pulls back. That is exactly what a spring does, so the response will be proportional to the push.

Balance and the polarizability

At equilibrium, the field pushing the nucleus equals the cloud's pull: $E = E_e$. So

$$E = \frac{1}{4\pi\epsilon_0}\frac{q\,d}{a^3} \quad\Longrightarrow\quad p = qd = (4\pi\epsilon_0 a^3)\,E.$$

The induced dipole $p = qd$ is proportional to E, which is Eq. 4.1 with $\alpha = 4\pi\epsilon_0 a^3$ (Eq. 4.2). The book says this is only accurate to a factor of about 4, because the real electron cloud is not a uniform ball.

Since $\frac{4}{3}\pi a^3 = v$ is the volume of the ball, $4\pi a^3 = 3v$, so $\alpha = 3\epsilon_0 v$. Atoms with bigger volume are easier to push, which makes sense.

Units check. $\mathbf{p}$ is in C·m and E is in N/C. So $\alpha = p/E$ is in C·m ÷ (N/C) = C²·m/N. Table 4.1 gives $\alpha/4\pi\epsilon_0$ in $10^{-30}\,\mathrm{m}^3$. $4\pi\epsilon_0$ has units C²/(N·m²), so multiplying by m^3 gives C²·m/N again. The two agree.

Molecules are less simple

For a molecule the push can depend on direction. CO_2 is easier to stretch along its axis than across it. So the field splits into a part along the axis and a part across it, and each part gets its own α:

$$\mathbf{p} = \alpha_\parallel \mathbf{E}_\parallel + \alpha_\perp \mathbf{E}_\perp.$$

If the molecule is very asymmetric, each component of $\mathbf{p}$ can depend on every component of $\mathbf{E}$. That is why there are nine numbers α_{ij} (Eq. 4.3). They form a 3×3 grid, the "tensor". Each row gives one component of $\mathbf{p}$: $p_x = \alpha_{xx}E_x + \alpha_{xy}E_y + \alpha_{xz}E_z$, and so on.

Problem 4.2: the electron cloud is not a ball

The hydrogen cloud has density $\rho(r) \propto e^{-2r/a}$. To find its field, use Gauss again, but now the enclosed charge is a sum over shells. For $r \ll a$, the exponential can be expanded as $e^{-2r/a} \approx 1 - 2r/a$. That approximation is what keeps the answer proportional to E. The idea is the same as before; only the charge distribution changed.

3. Permanent dipoles: the molecule already has one

Water is bent, with an angle of about $105°$. Oxygen pulls electrons toward itself, so one side is negative and the other is positive. It has a permanent dipole $\mathbf{p}$ even with no field. That is why water is a good solvent: it can pull at the charges of salt.

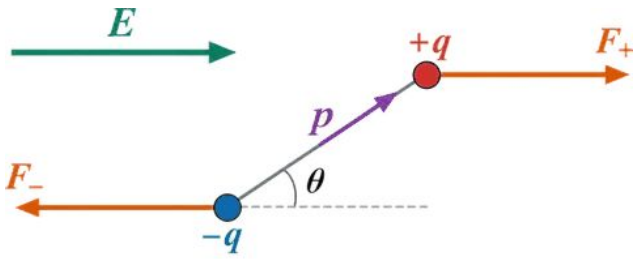
Torque: why it turns

Put a dipole in a uniform field. The positive end feels $q\mathbf{E}$ and the negative end feels $-q\mathbf{E}$. The net force is zero, so it does not move. But the forces are at different points, so they make a turning effect.

Take the centre as origin. The positive end is at $+\mathbf{d}/2$ and the negative end at $-\mathbf{d}/2$. The torque is

$$\mathbf{N} = \left(\tfrac{\mathbf{d}}{2} \times q\mathbf{E}\right) + \left(-\tfrac{\mathbf{d}}{2} \times (-q\mathbf{E})\right) = q\,\mathbf{d} \times \mathbf{E} = \mathbf{p} \times \mathbf{E}$$

Both terms point the same way, so they add. The direction of $\mathbf{p} \times \mathbf{E}$ tries to bring $\mathbf{p}$ in line with $\mathbf{E}$. That is why a polar molecule swings round to point with the field.



Net force zero; torque turns p toward E.

Figure 2. A permanent dipole in a uniform field. The two forces are equal and opposite, but they act at different points, so they turn the dipole.

Energy: where does $U = -\mathbf{p} \cdot \mathbf{E}$ come from?

Let the angle between $\mathbf{p}$ and $\mathbf{E}$ be θ. The torque from the field has size $pE\sin\theta$. To turn the dipole slowly from $90°$ to θ, I have to push against that torque. The work I do is

$$W = \int_{\pi/2}^{\theta} pE \sin\phi \, d\phi = pE\,(-\cos\theta) = -\mathbf{p}\cdot\mathbf{E}.$$

The work I do is stored as potential energy. So $U = -\mathbf{p}\cdot\mathbf{E}$, with the zero at $90°$. This is Problem 4.7, and it is the reason the dipole wants to line up: lowest energy is $\theta = 0$.

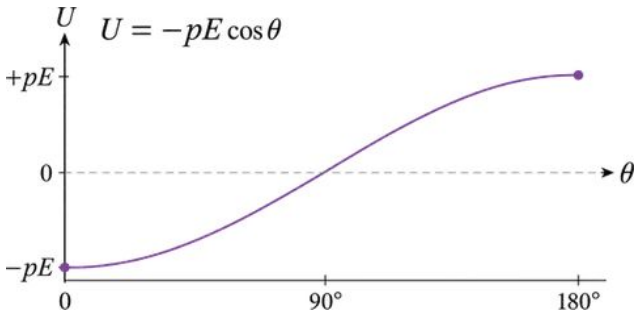


Figure 3. U against θ, with zero energy at 90°. The dipole is most stable at θ = 0, where U is lowest.

Why is it safe for a permanent dipole but not for an induced one? An induced dipole's size depends on E, so the simple $U = -\mathbf{p}\cdot\mathbf{E}$ does not capture the full energy (the book warns about this in footnote 3).

Non-uniform field: a force appears

In a non-uniform field, the two ends feel different fields. Let $\mathbf{d}$ be the separation. The field at the positive end is $\mathbf{E}(\mathbf{r} + \mathbf{d}/2)$, and at the negative end $\mathbf{E}(\mathbf{r} - \mathbf{d}/2)$. For a short dipole, the difference is a small change along $\mathbf{d}$:

$$\Delta\mathbf{E} \approx (\mathbf{d}\cdot\nabla)\mathbf{E}, \qquad \mathbf{F} = q\,\Delta\mathbf{E} = (\mathbf{p}\cdot\nabla)\mathbf{E}. \quad (4.5)$$

I did not know partial derivatives before this. ∇ just means "take the change in each direction". The idea is simple: a bigger change of E over the dipole length gives a bigger force. The book notes the field must change rather suddenly for this to matter inside a molecule, so it is usually small.

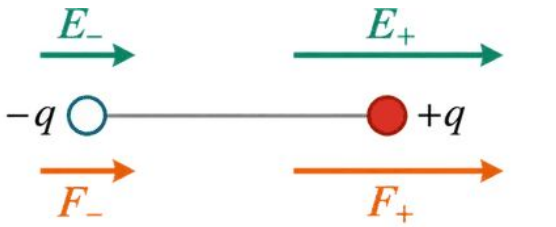


Figure 4. A short dipole in a non-uniform field. Each end feels its own field, and the stronger field at the + end gives the bigger force.

4. Polarization $\mathbf{P}$: many dipoles together

Now suppose each atom or molecule in a block has dipole $\mathbf{p}$, and there are N of them per unit volume, all pointing the same way. The total dipole in a small volume ΔV is $N\,\Delta V\,\mathbf{p}$. Dividing by the volume:

$$\mathbf{P} = \frac{\text{total dipole moment}}{\text{volume}} = N\mathbf{p}.$$

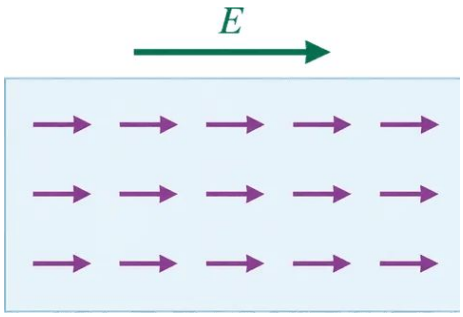


Figure 5. Aligned dipoles in a block. Polarization is the total dipole moment per unit volume.

Units: C·m per m³ gives C/m². $\mathbf{P}$ is the quantity that will matter later, because the material's own field depends on $\mathbf{P}$ and not on each atom separately.

Why the alignment is never perfect

A polar molecule has energy $-pE\cos\theta$ in the field. Thermal motion gives each molecule energy of order k_BT. Compare them: if $pE \ll k_BT$, thermal motion wins and the molecules point in all directions. At room temperature $k_BT \approx 0.026$ eV, and for a field of ordinary size the energy pE is tiny, so the alignment is weak. This is why the book says alignment is never complete and is lost almost at once when the field goes off.

The two mechanisms (induced and permanent) both give a net dipole along $\mathbf{E}$. Permanent dipoles turn;

induced ones stretch. That is the whole picture of §4.1.

Doubts I still have

- Why does the book say the uniform-ball model is only good to a factor of 4? (I think because the real cloud is more concentrated near the nucleus. Not yet checked.)
- Is ∇ the same as the gradient from my maths class? I need to revise partial derivatives before I can follow Eq. 4.5 fully.

Section 4.2 · The field of a polarized object

- Single dipole potential: $V = \dfrac{1}{4\pi\epsilon_0}\dfrac{\mathbf{p}\cdot\hat{\mathbf{r}}}{r^2}$ (Eq. 4.8).
- Polarization $\mathbf{P}$ is dipole moment per unit volume (§4.1.4).
- Maths you will need: gradient ∇, divergence $\nabla\cdot$, product rule, integration by parts, divergence theorem. Partial derivatives are beyond Plus Two, so revise them first.

1. Potential of a polarized object (§4.2.1)

1.1 Sum over the dipoles

Take a small piece $d\tau'$ at position $\mathbf{r}'$. It carries dipole moment $\mathbf{P}\,d\tau'$. Add up the dipole potentials of all the pieces:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int_V \frac{\mathbf{P}(\mathbf{r}')\cdot\hat{\boldsymbol{\mathcal{R}}}}{\mathcal{R}^2}\,d\tau' \quad (4.9)$$

Here $\boldsymbol{\mathcal{R}} = \mathbf{r} - \mathbf{r}'$ points from the source piece to the field point. This is correct, but the integral is awkward to evaluate directly.

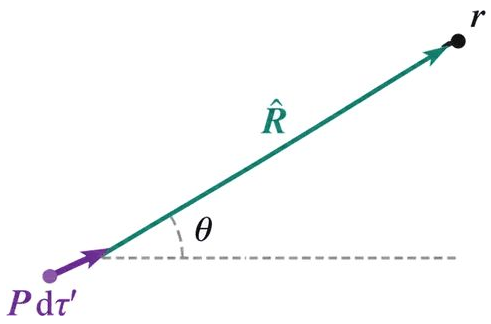


Figure 6. The geometry of Eq. 4.9.

1.2 Rewrite with a gradient

Differentiate with respect to the source point $\mathbf{r}'$ (not the field point):

$$\nabla'\left(\frac{1}{\mathcal{R}}\right) = \frac{\hat{\boldsymbol{\mathcal{R}}}}{\mathcal{R}^2}$$

So the integrand becomes a dot product of $\mathbf{P}$ with a gradient:

$$V = \frac{1}{4\pi\epsilon_0}\int_V \mathbf{P}\cdot\nabla'\left(\frac{1}{\mathcal{R}}\right)d\tau'$$

1.3 Integrate by parts

The product rule gives
$\nabla'\cdot\left(\dfrac{\mathbf{P}}{\mathcal{R}}\right) = \dfrac{1}{\mathcal{R}}\nabla'\cdot\mathbf{P} + \mathbf{P}\cdot\nabla'\left(\dfrac{1}{\mathcal{R}}\right)$.
Rearranging and splitting the integral:

$$V = \frac{1}{4\pi\epsilon_0}\left[\int_V \nabla'\cdot\left(\frac{\mathbf{P}}{\mathcal{R}}\right)d\tau' - \int_V \frac{1}{\mathcal{R}}(\nabla'\cdot\mathbf{P})\,d\tau'\right]$$

The divergence theorem turns the first integral into a surface integral:

$$V = \frac{1}{4\pi\epsilon_0}\oint_S \frac{1}{\mathcal{R}}\mathbf{P}\cdot d\mathbf{a}' - \frac{1}{4\pi\epsilon_0}\int_V \frac{1}{\mathcal{R}}(\nabla'\cdot\mathbf{P})\,d\tau'$$

1.4 Name the two pieces

DEFINITIONS

Surface: $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ (4.11), with $\hat{\mathbf{n}}$ the outward normal.

Volume: $\rho_b = -\nabla \cdot \mathbf{P}$ (4.12).

Substituting into Eq. 4.10:

RESULT

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \oint_S \frac{\sigma_b}{\mathcal{R}}\, da' + \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho_b}{\mathcal{R}}\, d\tau' \quad (4.13)$$

In words: a polarized object produces the same field as a volume charge ρ_b plus a surface charge σ_b. To find a field, first find these bound charges, then use Gauss's law or the usual potential integrals.

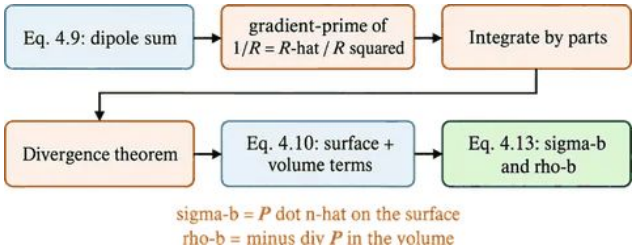


Figure 7. The derivation chain, Eq. 4.9 to Eq. 4.13.

2. Worked example: uniformly polarized sphere (Example 4.2)

- Sphere of radius R, uniform $\mathbf{P}$. Take the z-axis along $\mathbf{P}$.
- Volume charge: $\rho_b = -\nabla \cdot \mathbf{P} = 0$, because $\mathbf{P}$ is uniform.
- Surface charge: $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = P\cos\theta$.
- So we need the field of a $P\cos\theta$ surface charge on a sphere. The book uses the potential of that distribution (Eqs. 3.86–3.87):

$$V(r,\theta) = \begin{cases} \dfrac{P}{3\epsilon_0} r\cos\theta & r \le R \\ \dfrac{P}{3\epsilon_0}\dfrac{R^3}{r^2}\cos\theta & r \ge R \end{cases}$$

- Inside, $r\cos\theta = z$, so the field is uniform:

RESULT, INSIDE

$$\mathbf{E} = -\nabla V = -\frac{1}{3\epsilon_0}\mathbf{P} \text{ for } r < R \quad (4.14)$$

- Outside, the potential is the same as a perfect dipole at the centre, with the total dipole moment of the sphere:

RESULT, OUTSIDE

$$V = \frac{1}{4\pi\epsilon_0}\frac{\mathbf{p}\cdot\hat{\mathbf{r}}}{r^2} \text{ for } r \ge R \quad (4.15), \text{ with } \mathbf{p} = \tfrac{4}{3}\pi R^3 \mathbf{P} \quad (4.16)$$

Check: $\dfrac{(4/3)\pi R^3 P}{4\pi\epsilon_0} = \dfrac{R^3 P}{3\epsilon_0}$, which matches the outside potential.

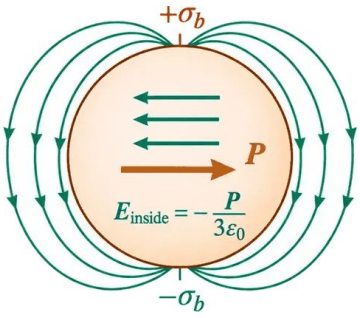


Figure 8. The uniformly polarized sphere: uniform field inside, dipole-like field outside.

3. Physical meaning of bound charge (§4.2.2)

3.1 A chain of dipoles

In a line of dipoles, the head of one cancels the tail of the next. Only the two ends are left, one positive and one negative. These are **bound charges**. They cannot be removed, because each electron stays on its own atom or molecule, but otherwise they are ordinary charges.

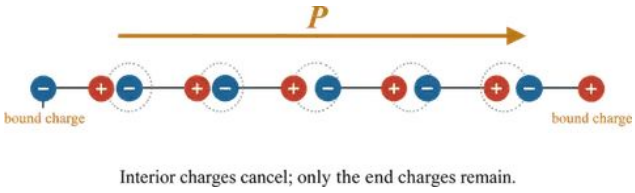


Figure 9. A chain of dipoles: interior charges cancel, end charges remain.

3.2 Amount at a flat end

Take a thin tube of dielectric parallel to $\mathbf{P}$, with cross-section A. A chunk of length d has dipole moment $P(Ad)$. Written as qd, the charge at the end is

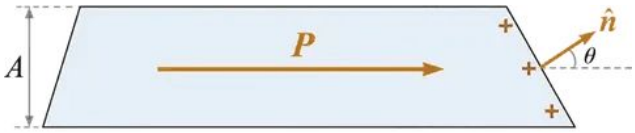
$$q = PA$$

For a cut perpendicular to $\mathbf{P}$, $\sigma_b = q/A = P$.

For an oblique cut at angle θ, the end area is $A_{\text{end}} = A/\cos\theta$. The charge is the same, so

> **RESULT**
> $$\sigma_b = \frac{q}{A_{\text{end}}} = P\cos\theta = \mathbf{P}\cdot\hat{\mathbf{n}}$$

This matches §4.2.1. Now we know where the bound charge comes from.



$$A_{\text{end}} = \frac{A}{\cos\theta}, \quad \sigma_b = P\cos\theta = \boldsymbol{P}\cdot\hat{\boldsymbol{n}}$$

Figure 10. An oblique cut: the same charge spread over a larger area.

3.3 Bound volume charge from a non-uniform P

If $\mathbf{P}$ changes from place to place, charge also builds up inside. Where $\mathbf{P}$ spreads out (diverges), the material loses more polarization out of a volume than it gains, leaving a net negative charge.

The total bound charge in a volume equals minus the flux of $\mathbf{P}$ out of it:

$$\int_V \rho_b\, d\tau = -\oint_S \mathbf{P}\cdot d\mathbf{a} = -\int_V (\nabla\cdot\mathbf{P})\, d\tau$$

This holds for every volume, so $\rho_b = -\nabla\cdot\mathbf{P}$ (4.12). This is the same result as §4.2.1.

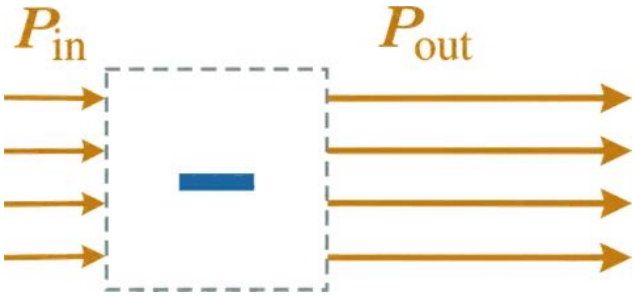


Figure 11. A small box: more P leaves than enters, so $\rho_b < 0$ inside.

3.4 Example 4.3: two overlapping spheres

- A uniformly polarized sphere is a positive sphere and a negative sphere of equal size, displaced by $\mathbf{d}$. Before polarization they cancel exactly.
- Polarization shifts them apart: a cap of positive charge appears at the top and a cap of negative charge at the bottom. These caps are σ_b.
- In the overlap, the field is $\mathbf{E} = -\frac{1}{4\pi\epsilon_0}\frac{q\mathbf{d}}{R^3}$ (from Prob. 2.19). With $\mathbf{p} = q\mathbf{d} = \frac{4}{3}\pi R^3\mathbf{P}$, this is $-\frac{1}{3\epsilon_0}\mathbf{P}$, which matches Eq. 4.14.
- Outside, each sphere acts as a point charge at its centre, giving a dipole potential, which matches Eq. 4.15.
- The displacement $\mathbf{d}$ is tiny, a small fraction of an atomic radius. The book's figure exaggerates it.

4. Macroscopic and microscopic fields (§4.2.3)

Background. The book says this subsection can be skipped without loss of continuity (footnote 5). It is not in the Module I syllabus list.

4.1 The problem

- Section 1 treated the atoms as point dipoles and smoothed them into a continuous $\mathbf{P}$.
- Outside the material, the molecules are far away, so this is safe.
- Inside, the true field varies enormously over atomic distances and changes with time. It is not practical to calculate.

4.2 The macroscopic field

The macroscopic field is the average of the true field over a region large enough to contain very many atoms, but small compared with the object. This is what is meant by "the field inside matter". It is the same kind of averaging that gives the density of a material.

4.3 Splitting the field

Draw a small sphere about the point $\mathbf{r}$. The average field is

$$\mathbf{E} = \mathbf{E}_{\text{out}} + \mathbf{E}_{\text{in}}$$

- $\mathbf{E}_{\text{out}}$: charges outside the sphere. Their average over the sphere equals the field at the centre (Prob. 3.51d). These dipoles are far away, so Eq. 4.9 applies (Eq. 4.17).
- $\mathbf{E}_{\text{in}}$: dipoles inside the sphere. Their average field depends only on the total dipole moment (Eq. 3.105):

$$\mathbf{E}_{\text{in}} = -\frac{1}{4\pi\epsilon_0}\frac{\mathbf{p}}{R^3} = -\frac{1}{3\epsilon_0}\mathbf{P} \quad (4.18)$$

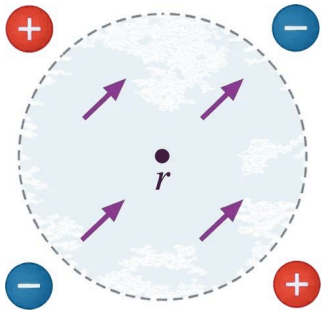


Figure 12. The averaging sphere: outside charges give E_{out}, inside dipoles give E_{in}.

4.4 Why this gives the same answer

$\mathbf{P}$ hardly changes across the small sphere, so the missing inside part is the centre field of a uniformly polarized sphere, $-\frac{1}{3\epsilon_0}\mathbf{P}$. Eq. 4.18 puts exactly this term back.

> **RESULT**
>
> $V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\mathbf{P}(\mathbf{r}')\cdot\hat{\boldsymbol{\mathcal{R}}}}{\mathcal{R}^2}\,d\tau'$ (4.19), over the whole dielectric

This is the integral of Eq. 4.9. So the method of §4.2.1 gives the macroscopic field inside the dielectric. The result does not depend on the shape of the averaging region, although the book uses a sphere.

5. What you should be able to do

- Derive Eq. 4.10 from Eq. 4.9.
- State σ_b, ρ_b and Eq. 4.13.
- Find the field inside and outside a uniformly polarized sphere (Eqs. 4.14–4.16).
- Explain $\sigma_b = \mathbf{P}\cdot\hat{\mathbf{n}}$ for an oblique cut.
- Explain why a diverging $\mathbf{P}$ gives negative bound charge.
- Show that the total bound charge of a neutral dielectric is zero (Prob. 4.14).

6. Problems not solved here

- Prob. 4.10: $\mathbf{P} = k\mathbf{r}$ in a sphere; find σ_b, ρ_b, and the fields.
- Prob. 4.11: frozen-in uniform P in a short cylinder (bar electret).
- Prob. 4.12: potential of the uniformly polarized sphere directly from Eq. 4.9.
- Prob. 4.13: long cylinder with P perpendicular to its axis.
- Prob. 4.14: prove the total bound charge vanishes.

7. Open points

- The outside field of the sphere (from Eq. 4.15) was checked through the potential only.
- Eq. 3.105 and Prob. 3.51d are cited from the book; their proofs are not repeated here.

Section 4.3 · The electric displacement

Book pp. 181–186. Pages 209 onward are outside this section.

Why a new field?

Inside a dielectric there are two kinds of charge: the free charge that someone put there, and the bound charge that comes from polarization. Gauss's law needs both, but we only control the free charge. So we look for a quantity that depends on the free charge alone.

1. Free and bound together

$$\rho = \rho_b + \rho_f \quad (4.20)$$

Put $\rho_b = -\nabla \cdot \mathbf{P}$ into Gauss's law: $\epsilon_0 \nabla \cdot \mathbf{E} = -\nabla \cdot \mathbf{P} + \rho_f$. Move the P term across and you get one divergence:

> **DEFINITION AND RESULT**
> $\mathbf{D} \equiv \epsilon_0 \mathbf{E} + \mathbf{P} \quad (4.21)$. Then
> $\nabla \cdot \mathbf{D} = \rho_f \quad (4.22)$, or in words
> $\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f,\text{enc}} \quad (4.23)$.

Now you only need the free charge to find D. Bound charge follows once the free charge is in place.

The surface is a special case. Right at the edge, the polarization drops to zero suddenly, so ρ_b becomes infinite. The book's way out: imagine a thin edge where P fades to zero. Then there is no surface term, and Gauss's law holds everywhere. The integral form (4.23) avoids the issue altogether.

2. Example 4.4: a wire with insulation

A long wire has uniform line charge λ and rubber insulation out to radius a. Draw a cylinder of radius s and length L. By symmetry D is radial. The free charge inside is λL:

> **RESULT**
> $D(2\pi s L) = \lambda L$, so $\mathbf{D} = \dfrac{\lambda}{2\pi s}\hat{\mathbf{s}} \quad (4.24)$. This holds inside and outside the insulation.

Outside, P is zero, so $\mathbf{E} = \mathbf{D}/\epsilon_0$. Inside, E cannot be found yet, because P is not known.

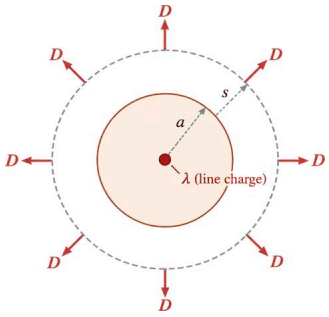


Figure 13. Wire with insulation and a cylindrical Gaussian surface (Example 4.4).

3. Careful: D is not just E with other charges

Eq. 4.22 looks like Gauss's law, so it is tempting to say "D is E, but sourced by free charge". The book says this is false. There is no Coulomb's law for D. A divergence does not fix a vector field; you need the curl too.

$$\nabla \times \mathbf{D} = \nabla \times \mathbf{P} \quad (4.25)$$

The curl of P is not always zero. Example: a bar electret with no free charge. If D came only from free charge, D would be zero everywhere, which would wrongly give $\mathbf{E} = -\mathbf{P}/\epsilon_0$ inside and zero outside. So the curl term matters.

Advice: check symmetry first. With spherical, cylindrical or plane symmetry, Gauss's law gives D directly, and the curl of P is automatically zero.

4. Boundary conditions

Across a boundary, the normal part of D jumps by the free surface charge (4.26). The tangential part of D jumps by the tangential part of P (4.27). This follows from the curl equation. For comparison, E jumps by the total surface charge over ϵ_0 in the normal direction (4.28), and its tangential part is continuous (4.29). Use the D versions when P is known.

5. Crystals: the choice is not unique (background)

In salt, each sodium ion can be paired with a chlorine ion above, below, left or right. Each choice gives a different P and a different split between free and bound charge. So the split is not uniquely defined for crystals. What is physical is the total charge, which is zero, and the change in P when a field is switched on.

Problems to try

- Prob. 4.15: a shell with $\mathbf{P} = (k/r)\hat{\mathbf{r}}$. Find E in three regions, by two methods.
- Prob. 4.16: fields at the centres of spherical, needle and wafer cavities.
- Prob. 4.17: sketch P, E and D for the bar electret.

Doubts

- Sign of the normal in Eqs. 4.26 and 4.28: not yet checked against §2.3.5.

Section 4.4 · Linear dielectrics

Book pp. 186–194 and 202–208. Pages 195–201 are not in the excerpt, so the first part of "forces on dielectrics" is missing here.

The big idea first

Up to now the book kept P and E separate. In this section it finally says: if the material is linear, P is just a fixed fraction of E. Everything after this is that one line, used in different situations.

1. The rule: P is proportional to E

For many materials, a weak field gives a polarization that grows in step with it:

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E} \quad (4.30)$$

χ_e is the susceptibility. The ϵ_0 is pulled out so that χ_e has no units. Its value depends on the material and on things like temperature.

A catch: E here is the **total** field, and part of it comes from the polarization itself. So you cannot just plug in the external field. The way out is to start from D, which only needs the free charge.

Footnote 10 says a strong field brings in a cubic term. Eq. 4.30 is only the first term of a series.

2. Where D and E become proportional

> **WORKING IT OUT**
> $\mathbf{D} = \epsilon_0 \mathbf{E} + \epsilon_0 \chi_e \mathbf{E} = \epsilon_0 (1 + \chi_e) \mathbf{E} \quad (4.31)$
> So $\mathbf{D} = \epsilon \mathbf{E}$ with $\epsilon = \epsilon_0 (1 + \chi_e) \quad (4.32, 4.33)$
> .
> The dielectric constant is
> $\epsilon_r = 1 + \chi_e = \epsilon / \epsilon_0 \quad (4.34)$.

- In vacuum there is no polarization, so $\epsilon = \epsilon_0$. The book does not like the name "permittivity of

free space", because it hides that vacuum is just a special case.

- For ordinary materials ϵ_r is bigger than 1, because P usually points the same way as E.
- The table of dielectric constants shows water at 80.1, silicon at 11.7, air at about 1.0005.

3. Example 4.5: a charged metal ball in a dielectric shell

A metal ball of radius a carries charge Q. Dielectric fills the space out to radius b. I need the potential at the centre, but I can't get E without P, and I can't get P without E. The way out: use the free charge Q, which I know, and Gauss's law for D:

$$\mathbf{D} = \frac{Q}{4\pi r^2}\hat{\mathbf{r}} \quad (r > a)$$

Then $\mathbf{E} = \mathbf{D}/\epsilon$ in the dielectric and $\mathbf{D}/\epsilon_0$ outside:

RESULT

$\mathbf{E} = \dfrac{Q}{4\pi\epsilon r^2}\hat{\mathbf{r}}$ for $a < r < b$, and

$\mathbf{E} = \dfrac{Q}{4\pi\epsilon_0 r^2}\hat{\mathbf{r}}$ for $r > b$.

Integrating gives the centre potential

$$V = \frac{Q}{4\pi}\left(\frac{1}{\epsilon_0 b} + \frac{1}{\epsilon a} - \frac{1}{\epsilon b}\right).$$

The bound charge explains why the field drops: the dielectric near the metal gets a negative surface layer at a (opposite to Q) and a positive one at b. A conducting shell would cancel the field completely; the dielectric cancels only part of it.

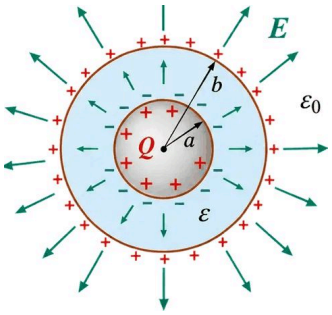


Figure 14. The charged ball and dielectric shell of Example 4.5.

4. Whole space filled with one dielectric

If one homogeneous linear dielectric fills all the space, the curl of P is zero, so D depends only on the free charge. The field is just the vacuum field divided by ϵ_r:

RESULT

$\mathbf{E} = \dfrac{1}{\epsilon_r}\mathbf{E}_{\text{vac}}$ (4.35), and for a point charge

$\mathbf{E} = \dfrac{1}{4\pi\epsilon}\dfrac{q}{r^2}\hat{\mathbf{r}}$ (4.36).

Coulomb's law is not wrong. The bound charge simply shields the charge.

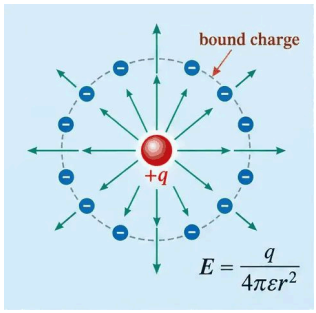


Figure 15. A point charge partly shielded by the bound charge around it.

5. Example 4.6: a capacitor with dielectric

The field is trapped between the plates. Filling them with dielectric divides E, and so V, by ϵ_r. For a fixed charge, the capacitance goes up by ϵ_r:

RESULT

$C = \epsilon_r C_{\text{vac}}$ (4.37). This is a common way to make a capacitor bigger.

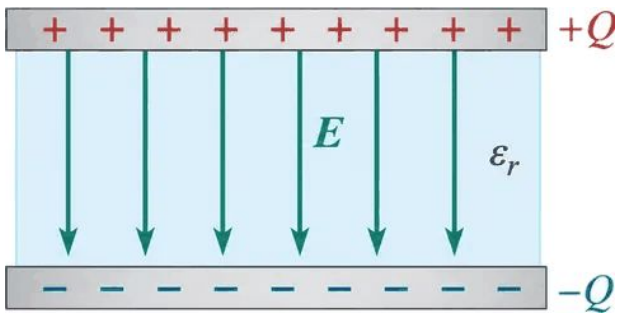


Figure 16. A parallel-plate capacitor filled with dielectric (Example 4.6).

6. Crystals: direction matters

Some crystals polarize more easily along some axes than others. Then each component of P depends on all three components of E, and there are nine susceptibilities (4.38). A material that behaves the same in every direction is called **isotropic**. When the book just says "linear dielectric", it means isotropic.

7. Boundary problems (background)

Inside a homogeneous linear dielectric, the bound charge is a fixed fraction of the free charge (4.39), so with no free charge inside, the potential obeys Laplace's equation. At a boundary, write the conditions in terms of free charge and potential:

$$\epsilon_{\text{above}} E^{\perp}_{\text{above}} - \epsilon_{\text{below}} E^{\perp}_{\text{below}} = \sigma_f \quad (4.40)$$

$$V_{\text{above}} = V_{\text{below}} \quad (4.42)$$

Example 4.7 is the dielectric sphere in a uniform field. It is like the conducting sphere of Ex. 3.8, but the cancellation is only partial:

- The three conditions: continuous V at $r = R$; $\epsilon\, \partial V_{\text{in}}/\partial r = \epsilon_0\, \partial V_{\text{out}}/\partial r$ at $r = R$; and $V_{\text{out}} \to -E_0 r \cos\theta$ far away.
- Expand inside and outside in Legendre polynomials, match term by term (only $l = 1$ survives), and get

RESULT
$A_1 = -\dfrac{3}{\epsilon_r + 2} E_0$, and inside $\mathbf{E} = \dfrac{3}{\epsilon_r + 2}\mathbf{E}_0 \quad (4.49)$. The field inside is uniform, though weaker than $\mathbf{E}_0$.

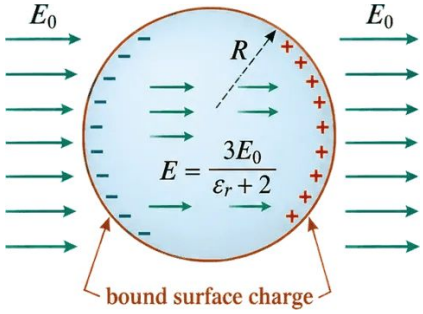


Figure 17. Dielectric sphere in a uniform field: the field inside is uniform and weaker.

8. Force on a dielectric slab (partly in excerpt)

A slab is pulled into a charged capacitor. The fringing fields that pull it are hard to work out, so use energy: the force is minus the change of energy per unit displacement. At fixed charge, $W = \frac{1}{2}Q^2/C$, so

RESULT
$F = -\dfrac{dW}{dx} = \frac{1}{2}V^2\dfrac{dC}{dx} \quad (4.64)$, and with $dC/dx = -\epsilon_0\chi_e w/d$:
$F = -\dfrac{\epsilon_0\chi_e w}{2d}V^2 \quad (4.65)$.

- The minus sign: the slab is pulled **into** the capacitor.
- Careful with the sign. If you hold V fixed and use $\frac{1}{2}CV^2$, you get the wrong sign. With a battery, the work $V\,dQ$ has to be added (4.66), and then the answer comes out the same as (4.64).
- The force depends only on the charges present at that moment, not on whether Q or V is held fixed.

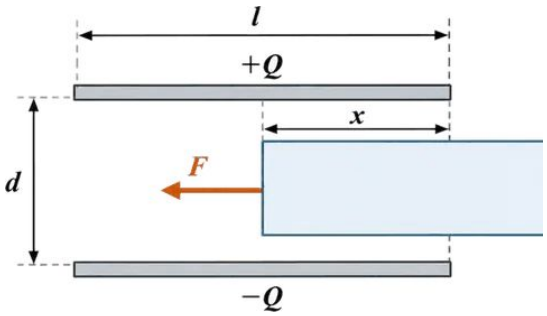


Figure 18. The slab is pulled into the capacitor (Eqs. 4.59–4.65).

9. Why the simple formula for χ is not quite right (background)

The naive guess is $\chi_e = N\alpha/\epsilon_0$, with N atoms per volume (4.70). But α was defined for an atom in an outside field, while the field that acts on each atom is the total field, which includes the others. Averaging over a small sphere around each atom gives $\mathbf{E} = (1 - N\alpha/3\epsilon_0)\mathbf{E}_{\text{else}}$ (4.71). Solving gives the Clausius–Mossotti formula:

RESULT

$\chi_e = \dfrac{N\alpha/\epsilon_0}{1 - N\alpha/3\epsilon_0}$, equivalently $\alpha = \dfrac{3\epsilon_0}{N}\dfrac{\epsilon_r - 1}{\epsilon_r + 2}$ (4.72).

For polar molecules, the Boltzmann factor $e^{-u/kT}$ with $u = -pE\cos\theta$ gives the Langevin formula (4.73), $P = Np[\coth(pE/kT) - kT/pE]$. For $pE \ll kT$ the medium is linear.

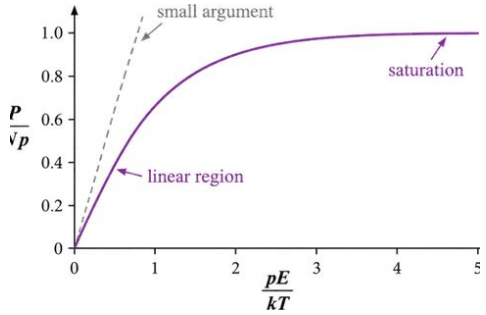


Figure 19. The Langevin function: linear for small pE/kT, saturating for large.

10. Problems to try

- Prob. 4.18: two slabs between plates. Find D, E, P, V and all the bound charge.
- Prob. 4.20 and 4.21: sphere with uniform free charge; coaxial cable half-filled.
- Prob. 4.28: oil rising between tubes (uses 4.64).
- Prob. 4.38 and 4.37: dipole in a dielectric sphere; field lines bending at an interface (4.68).
- Prob. 4.42–4.44: Clausius–Mossotti and Langevin.

Doubts I still have

- Pages 195–201 are not in my excerpt, so I don't have the start of the force section.
- I haven't worked the Langevin small-argument step (the one that gives $\chi_e \approx Np^2/3\epsilon_0 kT$) on paper yet.

Section 6.1 · Magnetization

Book pp. 268–276. Pages 209–267 are not in the excerpt.

Big picture

The book opens with a surprise: magnets are not a separate thing from moving charges. Inside an atom there are tiny currents, electrons going round the nucleus and spinning. Each one acts like a small magnet, which the book calls a magnetic dipole. Normally they point every which way and cancel. A field lines them up a little, and the material is then **magnetized**.

1. Three kinds of response

- **Paramagnet:** magnetization points along B.
- **Diamagnet:** magnetization points against B.
- **Ferromagnet:** keeps its magnetization after the field is off. Its state depends on its history, so the book leaves it to the end.

The book warns that the classical picture is incomplete. Real electron spin is a fixed point dipole, not a spinning ball.

2. Torque on a current loop

Take a rectangular loop of sides a and b, carrying current I, in a uniform field B. The forces on the sloping sides cancel and do not turn the loop. The forces on the horizontal sides are equal and opposite, but they are a little apart, so they make a turning effect:

$$\mathbf{N} = IabB\sin\theta\,\hat{\mathbf{x}} = \mathbf{m}\times\mathbf{B} \quad (6.1)$$

This is the magnetic version of $\mathbf{p}\times\mathbf{E}$ from §4.1. The loop turns to line up with B. Any loop can be cut into little rectangles, so the result is general.

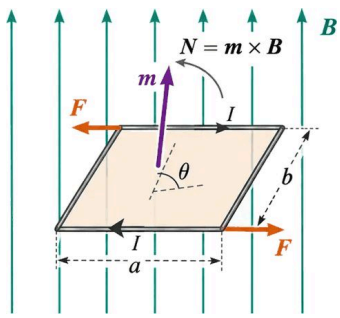


Figure 20. Loop in a uniform field: the horizontal sides give the torque.

3. Why paramagnetism is usually weak

Electrons pair up with opposite spins, and the pairs cancel each other's dipoles. So it is mostly atoms with an odd number of electrons that feel the torque. Heat then upsets the alignment, so it is never complete.

4. Force on a magnet

- In a uniform field a loop feels no net force, because $\oint d\mathbf{l} = 0$.
- In a non-uniform field a ring above a solenoid feels a force $F = 2\pi IRB\cos\theta$ (6.2).
- For a tiny dipole: $\mathbf{F} = \nabla(\mathbf{m}\cdot\mathbf{B})$ (6.3). This is the same form as $\nabla(\mathbf{p}\cdot\mathbf{E})$.

Two models are in use. The Gilbert model has separate north and south "charges", which makes the algebra easy. The Ampère model uses current loops, which is the real one, because magnetic monopoles do not exist. Use Gilbert for a quick picture, and never for exact close-range answers.

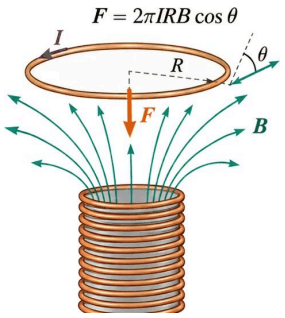


Figure 21. Ring above a solenoid: the net force in the fringing field (6.2).

5. Orbiting electrons and diamagnetism

Model an electron as a circle of radius R and speed v. Its current is $I = -ev/2\pi R$, and its dipole is

$$\mathbf{m} = -\tfrac{1}{2}evR\,\hat{\mathbf{z}} \quad (6.4)$$

Without a field, the electric pull alone supplies the centripetal force (6.5). With a field B perpendicular to the orbit, the Lorentz force $-e\mathbf{v}\times\mathbf{B}$ adds to it (6.6). Subtracting (6.5) from (6.6) and taking the change to be small:

RESULT

$\Delta v = \dfrac{eRB}{2m_e}$ (6.7). The electron speeds up.

A faster electron has a changed dipole:

RESULT

$\Delta \mathbf{m} = -\dfrac{e^2R^2}{4m_e}\mathbf{B}$ (6.8). The change is against B, whichever way the electron goes round.

Every atom gets a small extra dipole against B. That is diamagnetism. It exists in all atoms but is weaker than paramagnetism, so it shows up mostly in atoms with an even number of electrons.

The book says this classical model is flawed, and that diamagnetism is really quantum. Also, the speed change is caused by the induced electric field of a changing B (Chapter 7), not by B doing work.

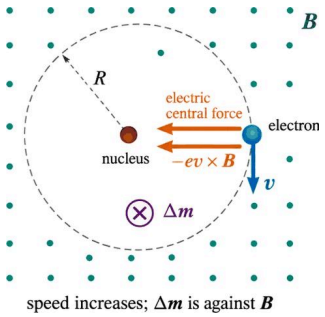


Figure 22. The electron orbit speeds up when B is turned on.

6. Magnetization M

DEFINITION

$\mathbf{M} \equiv$ magnetic dipole moment per unit volume (6.9). It plays the role P played for dielectrics.

The book takes M as given and asks what field it makes (next section). It does not matter whether M comes from spin, from orbits, or from ferromagnetism.

Forces on samples in a non-uniform field are very small. Paramagnets are drawn in; diamagnets are pushed out. Iron feels 10^4 to 10^5 times more than a comparable sample of a weak material, which is why we could ignore magnetization in Chapter 5.

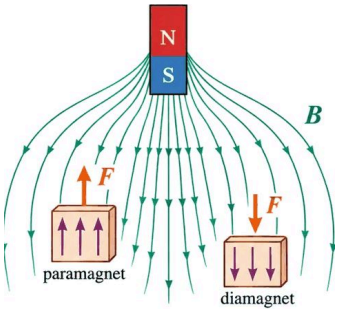


Figure 23. Paramagnetic sample drawn in; diamagnetic sample pushed out.

7. Problems to try

- Prob. 6.2: show the torque on any steady current in B is $\mathbf{m} \times \mathbf{B}$.
- Prob. 6.4: derive $\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B})$ for a small square loop.
- Prob. 6.6: which materials are para- and diamagnetic.

Doubts

- I have not checked Hund's rule (footnote 8) yet.
- Eq. 6.8 assumes the orbit stays circular. The book says that is not justified in general, so I should not use it for exact numbers.

Section 6.2 · The field of a magnetized object

Book pp. 276–281.

The idea

Last section we found that a magnetized lump is a bunch of tiny loops. This section asks: can we replace all those loops with simpler currents on the surface and inside, and get the same field? Yes. That is what the book means by **bound currents**.

1. Summing the loops

Each tiny piece of the object is one dipole, $\mathbf{M}\,d\tau'$. The field of one dipole is in terms of its vector potential (Eq. 6.10). Adding them all up gives (6.11). The integral is messy, so we rewrite it.

The trick is the identity $\boldsymbol{\nabla}'(1/\mathcal{R}) = \hat{\boldsymbol{\mathcal{R}}}/\mathcal{R}^2$. Using integration by parts and the divergence theorem, the messy integral splits into two simpler ones (6.12). One is over the volume, one over the surface.

> **TWO NEW CURRENTS**
> Inside: $\mathbf{J}_b = \boldsymbol{\nabla} \times \mathbf{M}$ (6.13). On the surface: $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$ (6.14).

> **RESULT**
> The field of the magnetized object is the same as the field of $\mathbf{J}_b$ inside and $\mathbf{K}_b$ on the surface (6.15). This is the magnetic version of the bound charge trick in §4.2.

2. Example 6.1: magnetized sphere

Let M point along z. Inside, $\boldsymbol{\nabla} \times \mathbf{M} = \mathbf{0}$, so there is no volume current. On the surface, $K_b = M \sin\theta$, flowing around the sphere. That is exactly the current on a spinning uniformly charged shell. So the magnetized sphere is equivalent to a spinning shell.

> **RESULT**
> Inside, $\mathbf{B} = \frac{2}{3}\mu_0\mathbf{M}$ (6.16), which is uniform. Outside, it is a perfect dipole with $\mathbf{m} = \frac{4}{3}\pi R^3\mathbf{M}$.

The electric version (uniformly polarized sphere) also gives a uniform field inside, but its factor is $-\frac{1}{3}$. The factors differ, but the shape of the answer matches.

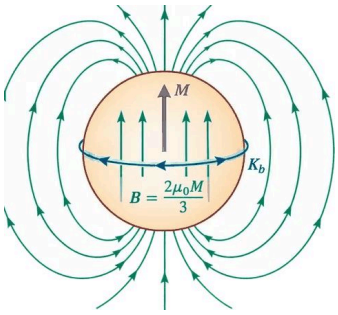


Figure 24. Magnetized sphere: uniform field inside, dipole field outside.

3. Why these currents are real

Picture a slab made of tiny loops all pointing the same way. Inside, each loop's right side cancels its neighbour's left side. At the edge there is no neighbour, so one current is left running around the boundary. Its strength is $K_b = M$, because one loop of area a and thickness t has $m = Mat = Ia$, so $I = Mt$.

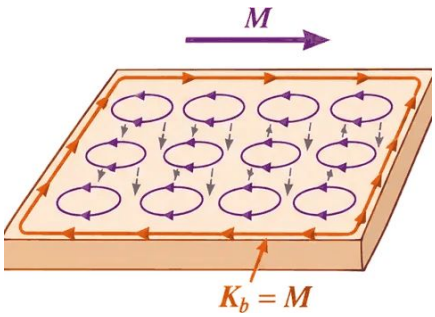


Figure 25. Internal currents cancel; the edge carries the bound current.

If M varies from place to place, the cancellation fails inside too. The result is $\mathbf{J}_b = \boldsymbol{\nabla} \times \mathbf{M}$. Check: $\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{M}) = 0$, so this current is steady, as it should be.

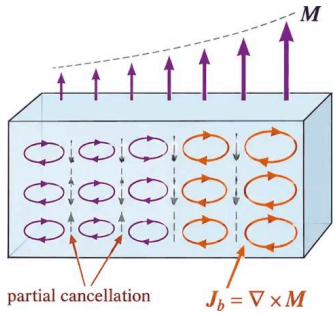


Figure 26. Non-uniform M: some internal currents survive.

Note: each electron only moves within its own atom. The current is "bound" to the atoms, but over the whole object it is a real current.

4. Inside matter

The true field inside matter changes wildly from place to place. The field we use is the average over many atoms. The methods above give that average. Prob. 6.11 asks for the proof.

5. Problems to try

- Prob. 6.7: infinite cylinder with M along its axis.
- Prob. 6.9: short cylinder (bar magnet). Compare the bar electret, Prob. 4.11.

Doubts

- I should check the sign of $\mathbf{K}_b$ against Example 5.9.

Section 6.3 · The auxiliary field H

Book pp. 281–286.

Why do we need a new field?

Ampère's law (Chapter 5) says the curl of B equals μ_0 times the current. But now there are two kinds of current: the free current from the wires, and the bound current from the magnetization. We want a rule that only mentions the current we control.

1. Splitting the current

The total current is $\mathbf{J} = \mathbf{J}_b + \mathbf{J}_f$ (6.17). The free current comes from the battery. The bound current comes from the atoms lining up. Put $\mathbf{J}_b = \nabla \times \mathbf{M}$ into Ampère's law and collect the curls:

$$\nabla \times \left(\frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \right) = \mathbf{J}_f$$

The bracket is called **H**:

DEFINITION AND RESULT

$\mathbf{H} \equiv \frac{1}{\mu_0}\mathbf{B} - \mathbf{M}$ (6.18), and then

$\nabla \times \mathbf{H} = \mathbf{J}_f$ (6.19), or

$\oint \mathbf{H} \cdot d\mathbf{l} = I_{f,\text{enc}}$ (6.20).

This is the magnetic twin of D. Gauss's law used D so that you only need the free charge; Ampère's law uses H so that you only need the free current. Practical people prefer H, because they set a current on a dial and not a field.

2. Example 6.2: a wire with current

Take a long rod of copper with a uniform current I. Copper is weakly diamagnetic, so the bound currents oppose I inside and follow I on the surface. The field is azimuthal, and a loop of radius s inside the rod encloses the fraction s^2/R^2 of the current:

RESULTS

Inside: $\mathbf{H} = \frac{I}{2\pi R^2} s\,\hat{\boldsymbol{\phi}}$ (6.21).

Outside: $\mathbf{H} = \frac{I}{2\pi s}\hat{\boldsymbol{\phi}}$ (6.22). Here $\mathbf{B} = \mu_0\mathbf{H}$, the same as for an ordinary wire.

Inside, I cannot get B, because I do not know M yet.

{{FIG:s63-rod-H}}

Figure 27. H inside the rod rises with s, and outside it falls as 1/s.

3. Two traps

- Eq. 6.19 looks like Ampère's law, but a curl alone does not decide a field. You also need the divergence.
- Unlike B, the divergence of H is not zero: $\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}$ (6.23).
- Bar magnet: no free current at all, so a careless use of (6.20) gives $\mathbf{H} = \mathbf{0}$. That is wrong. The divergence of M at the ends is what makes H non-zero there.

Rule: look for symmetry first. Only then trust (6.20) on its own.

4. Boundary conditions (beyond our list)

Across a boundary, H's normal part jumps by the negative of M's jump (6.24), and its tangential part jumps by the free surface current (6.25). The book suggests checking them against Ex. 6.2 and Prob. 6.14.

5. Problems to try

- Prob. 6.12: cylinder with $\mathbf{M} = ks\,\hat{\mathbf{z}}$. Do it by bound currents, then by (6.20).
- Prob. 6.14: sketch M, B, H for the bar magnet.

Doubts

- Prob. 6.15 (H as the gradient of a potential) I have not tried yet.

Section 6.4 · Linear and nonlinear media

Book pp. 286–294.

The rule for most materials

For most materials, M is proportional to the field, as long as the field is not too strong. Writing it in terms of B is correct but awkward, so the book uses H:

> **RULE**
> $\mathbf{M} = \chi_m \mathbf{H} \quad (6.29)$. The number χ_m is the magnetic susceptibility. It is positive for paramagnets and negative for diamagnets. Typical size: 10^{-5}.

Then B follows: $\mathbf{B} = \mu_0(1 + \chi_m)\mathbf{H}$ (6.30). Define the permeability $\mu = \mu_0(1 + \chi_m)$, so $\mathbf{B} = \mu\mathbf{H}$ (6.31, 6.32). In vacuum $\chi_m = 0$, so $\mu = \mu_0$.

Table 6.1 (book): magnetic susceptibilities.

Diamagnetic	χ_m	Paramagnetic	χ_m
Bismuth	-1.7×10^{-4}	Oxygen (O_2)	1.7×10^{-6}
Gold	-3.4×10^{-5}	Sodium	8.5×10^{-6}
Silver	-2.4×10^{-5}	Aluminum	2.2×10^{-5}
Copper	-9.7×10^{-6}	Tungsten	7.0×10^{-5}
Water	-9.0×10^{-6}	Platinum	2.7×10^{-4}
Carbon dioxide	-1.1×10^{-8}	Oxygen (liquid, −200 °C)	3.9×10^{-3}
Hydrogen (H_2)	-2.1×10^{-9}	Gadolinium	4.8×10^{-1}

Example 6.3: solenoid with a core

An infinite solenoid with n turns per metre and current I. H comes from the free current alone, so $\mathbf{H} = nI\hat{\mathbf{z}}$. Then $\mathbf{B} = \mu_0(1 + \chi_m)nI\hat{\mathbf{z}}$. A paramagnetic core makes B bigger, a diamagnetic core makes it smaller. The reason: the bound surface current runs along I for $\chi_m > 0$ and against it for $\chi_m < 0$.

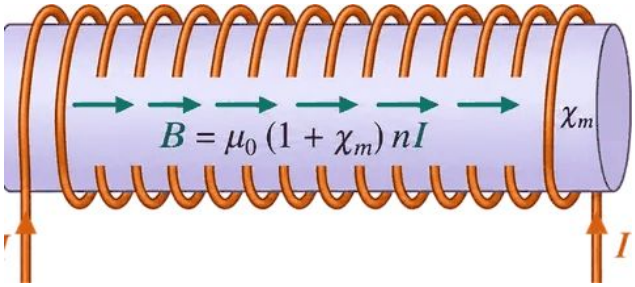


Figure 28. Solenoid with a linear core.

Why M can have a divergence in a linear medium

For the electric field we said B has zero divergence, and that is always true. But M can have a divergence at a boundary between two materials. Take a pillbox across the boundary: M is zero on one side and not on the other, so the flux through the box is not zero. So the divergence of M is not zero there.

Inside a homogeneous linear material, the bound current is a fixed fraction of the free current (6.33): $\mathbf{J}_b = \chi_m \mathbf{J}_f$. So unless free current flows through the material, the bound current sits on the surface.

Ferromagnets are not linear

A ferromagnet keeps its magnetization with no field. Its alignment is frozen in. In a ferromagnet, neighbouring dipoles like to point the same way. This is quantum, and the book does not explain it. So for a ferromagnet, the susceptibility is not a constant.

Domains

Iron is not a magnet in everyday life, because the aligned regions (domains) point in random directions. A field grows the domains that point along it, by moving the walls between them. If the field is strong enough, one domain takes over. That is saturation. When the field is switched off, most of the domains stay where they are, so you have a permanent magnet.

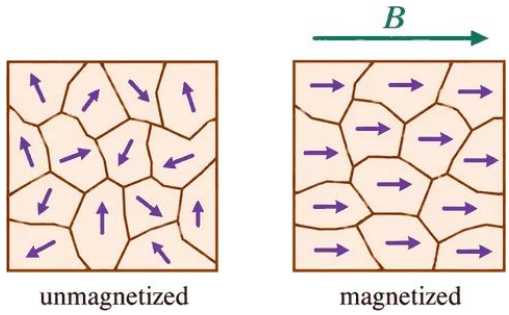


Figure 29. Domains: aligned patches, growing when the field moves their walls.

Hysteresis: why history matters

Run a current through a coil around a piece of iron. Follow the path: from zero, to saturation (b), back to zero current with some magnetization left (c), reversed current until M is zero (d), reversed saturation (e), back to zero with reversed residual (f), and so on. The closed path is the hysteresis loop. Notice that zero current does not mean zero magnetization: at a, c and f the current is zero but M differs.

Plots usually use B against H. For a long solenoid H = nI, and B is almost $\mu_0 M$ because M is much bigger than H in practice.

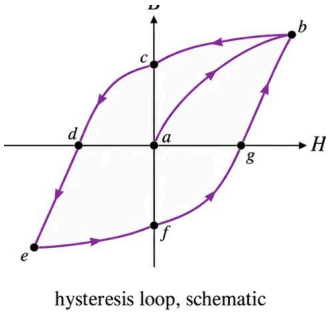


Figure 30. Hysteresis loop (schematic).

Curie point

Heat destroys the alignment. For iron the threshold is 770 °C. Below it, iron is ferromagnetic. Above it, iron is only paramagnetic. The change is sudden, like water boiling.

Problems to try

- Prob. 6.17: a wire of linear material. Field at radius s and total bound current.
- Prob. 6.19: estimate χ for copper and compare with Table 6.1.
- Prob. 6.21: energy of a dipole, $U = -\mathbf{m} \cdot \mathbf{B}$ (6.34).
- Prob. 6.28: how field lines bend at a boundary between magnetic materials.

Doubts

- I have not checked the Table 6.1 numbers against another source.